Bookmark 1 : High Prior Visits | Bookmark 2 : Extended Stay Risk | Bookmark 3 : Discharge Risk | Bookmark 4 : Med Change Impact | Bookmark 5 : Reset Filters

Readmission Analysis – All Diagnosis Groups

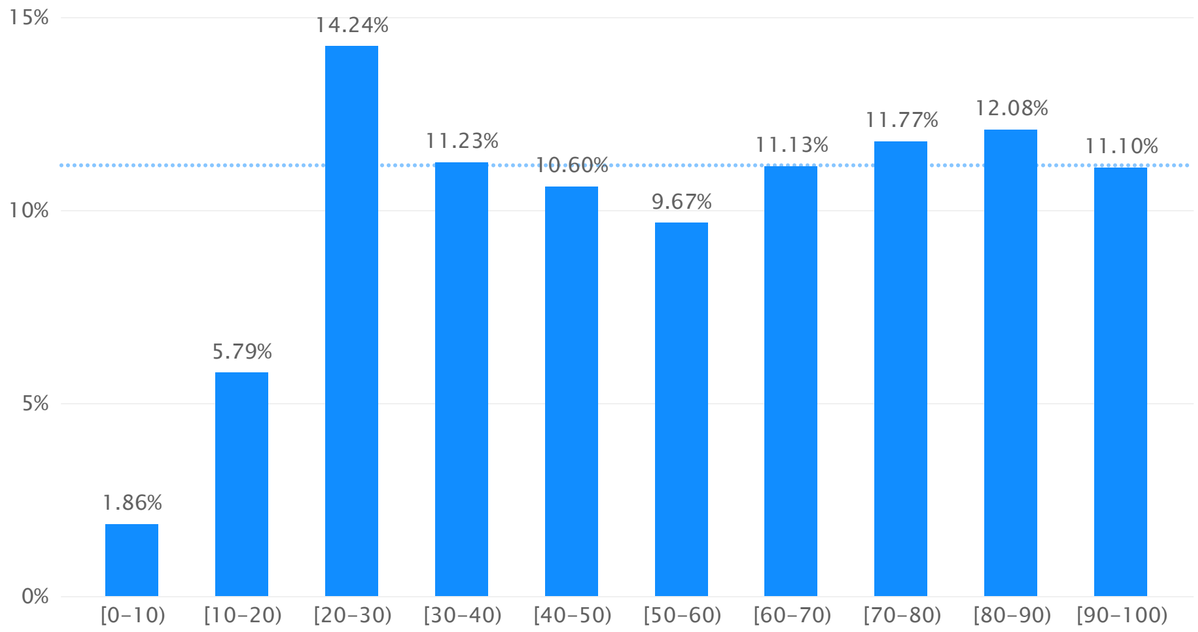
Primary Diagnosis Group: All

Medication Change: All

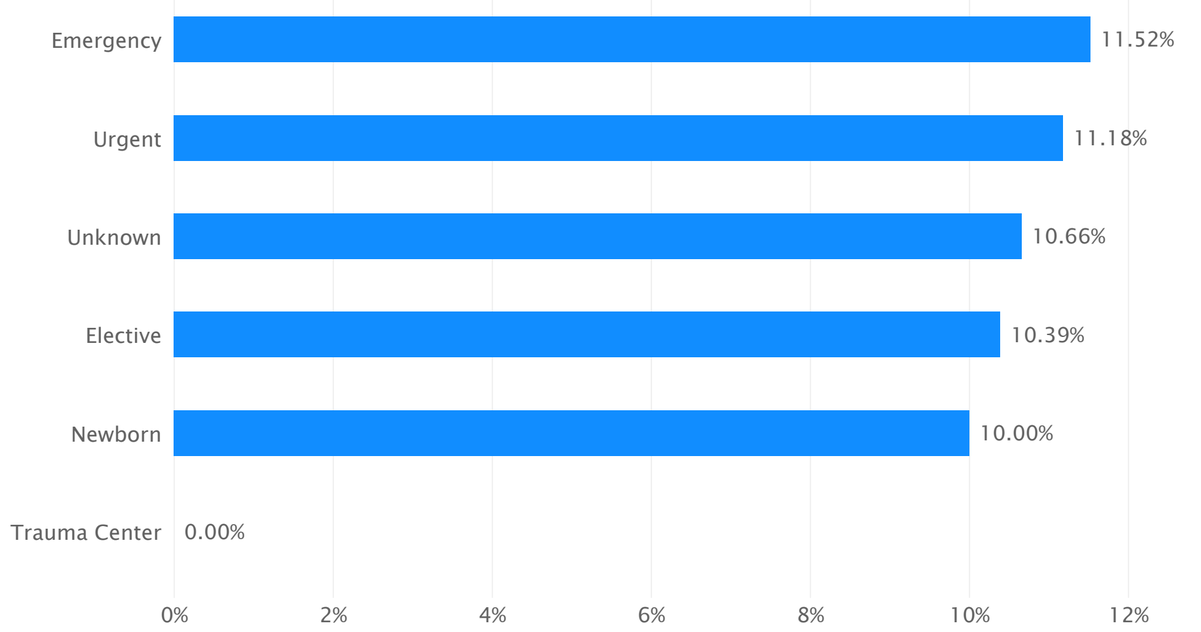
Length of Stay Band: All

Total Encounters	Readmission Rate (30D) %	Avg Length of Stay (Days)	Avg Medications
102K	11.16%	4.40	16.02

Readmission Rate (30D) %
by Age Group



Readmission Rate (30D) %
by Admission Type



Bookmark 1 : High Prior Visits | Bookmark 2 : Extended Stay Risk | Bookmark 3 : Discharge Risk | Bookmark 4 : Med Change Impact | Bookmark 5 : Reset Filters

Readmission Analysis – All Diagnosis Groups

Primary Diagnosis Group: All

Medication Change: All

Length of Stay Band: All

Readmission Rate (Med Changed) %

11.82%

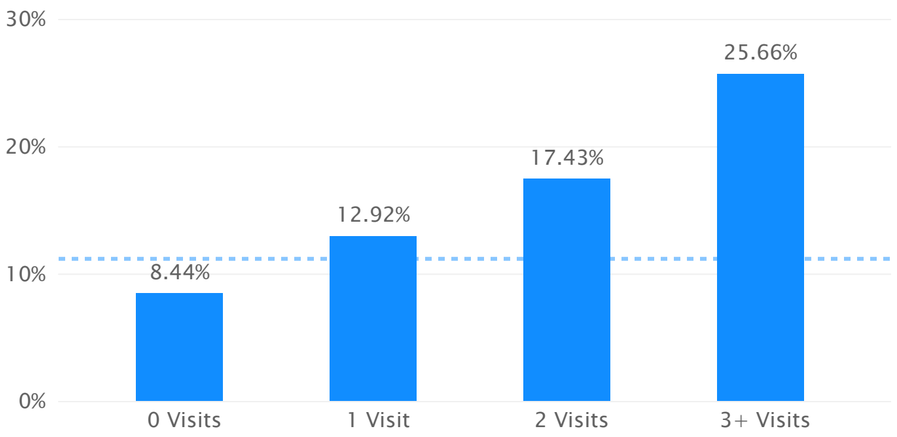
Readmission Rate (30D) %

11.16%

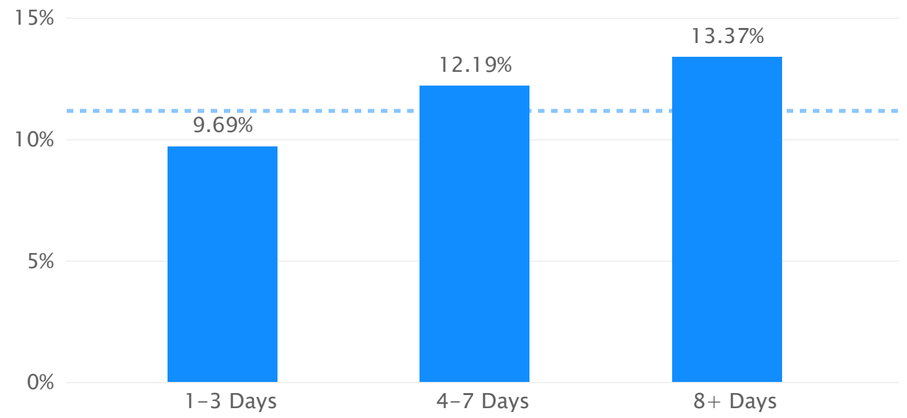
High Risk Encounters

1.43K

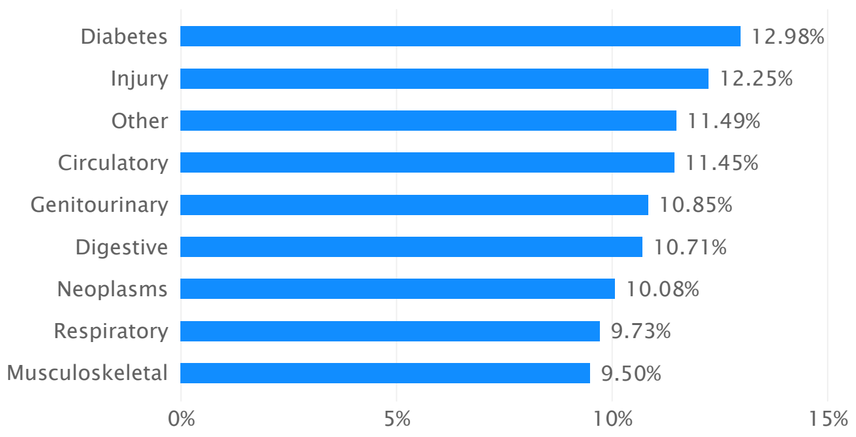
Readmission Rate (30D) %
by Prior Inpatient Visits



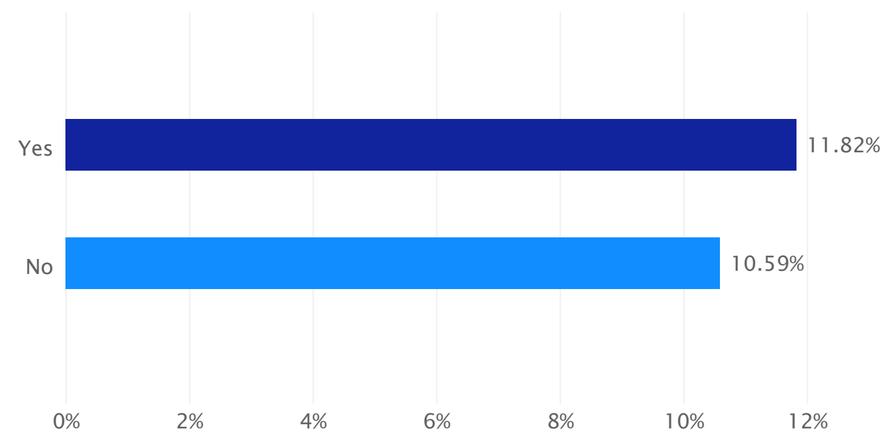
Readmission Rate (30D) %
by Length of Stay Band



Readmission Rate (30D) %
by Primary Diagnosis Group



Readmission Rate (30D) %
by Medication Change



Bookmark 1 : High Prior Visits | Bookmark 2 : Extended Stay Risk | Bookmark 3 : Discharge Risk | **Bookmark 4 : Med Change Impact** | Bookmark 5 : Reset Filters

Readmission Analysis – All Diagnosis Groups

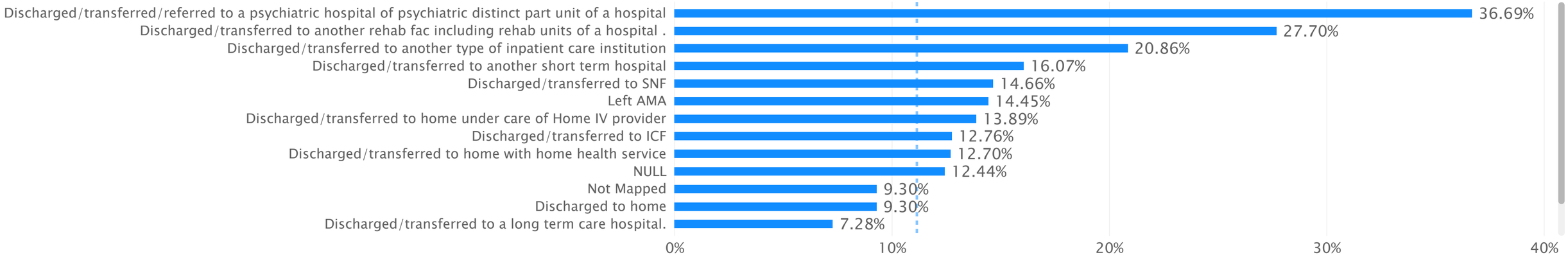
Primary Diagnosis Group: All

Medication Change: All

Length of Stay Band: All

Discharge Disposition	Admitted as an inpatient to this hospital		Discharged to home		Discharged/transferred to a federal health care facility.		Discharged/transferred to a long term care hospital.		Discharged/t
Age Group	Readmission Rate (30D) %	Total Encounters	Readmission Rate (30D) %	Total Encounters	Readmission Rate (30D) %	Total Encounters	Readmission Rate (30D) %	Total Encounters	Readmission
[0-10)			2.07%	145					
[10-20)	0.00%	1	5.76%	625					
[20-30)			13.81%	1405			0.00%	3	
[30-40)			10.42%	3081					
[40-50)	50.00%	2	9.45%	7448			3.70%	27	
[50-60)			7.96%	12252	0.00%	3	10.20%	49	
[60-70)	45.45%	11	9.39%	14437	0.00%	1	8.16%	98	
[70-80)	33.33%	3	9.39%	13689			8.26%	109	
[80-90)	50.00%	4	9.90%	6444	0.00%	1	4.90%	102	
Total	**42.86%**	**21**	**9.30%**	**60234**	**0.00%**	**5**	**7.28%**	**412**	

Readmission Rate (30D) %
by Discharge Disposition



Readmission Analysis – All Diagnosis Groups

Primary Diagnosis G... All | Medication Change All | Length of Stay Band All

Age Group	Diagnosis Group	Time in Hospital (Days)	Medications	Lab Procedures	Medication Change	Readmitted Status
[50-60)	Circulatory	1	1	30	No	NO
[50-60)	Circulatory	1	1	34	No	NO
[50-60)	Circulatory	1	1	39	Yes	NO
[50-60)	Circulatory	1	1	42	No	>30
[50-60)	Circulatory	1	1	44	No	NO
[50-60)	Circulatory	1	1	47	No	NO
[50-60)	Circulatory	1	1	53	No	NO
[50-60)	Circulatory	1	2	4	No	NO
[50-60)	Circulatory	1	2	10	Yes	NO
[50-60)	Circulatory	1	2	21	No	NO
[50-60)	Circulatory	1	2	31	No	NO
[50-60)	Circulatory	1	2	32	No	NO
[50-60)	Circulatory	1	2	35	No	NO
[50-60)	Circulatory	1	2	40	No	>30
[50-60)	Circulatory	1	2	41	No	NO
[50-60)	Circulatory	1	2	42	No	>30
[50-60)	Circulatory	1	2	43	No	>30
[50-60)	Circulatory	1	3	1	No	NO
[50-60)	Circulatory	1	3	2	No	NO
[50-60)	Circulatory	1	3	3	No	>30
[50-60)	Circulatory	1	3	8	No	NO
[50-60)	Circulatory	1	3	21	No	<30
[50-60)	Circulatory	1	3	33	No	NO
[50-60)	Circulatory	1	3	34	No	NO
[50-60)	Circulatory	1	3	40	No	NO
[50-60)	Circulatory	1	3	41	No	NO
[50-60)	Circulatory	1	3	44	Yes	>30
Total						

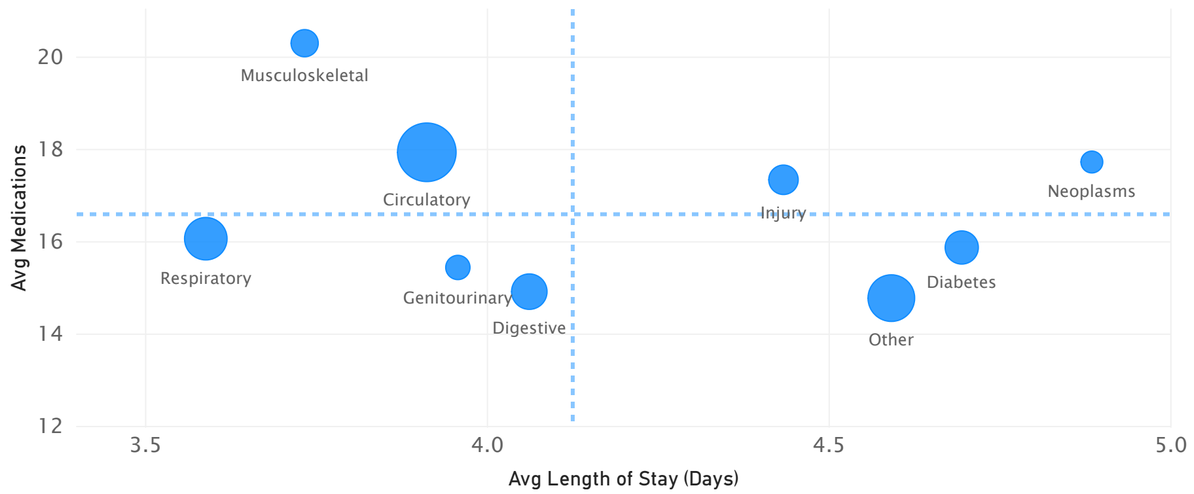
High-Risk Segment Share %
1.38%

Avg Length of Stay (Days)
4.13

High Risk Encounters
238

Readmission Rate (30D) %
9.67%

Avg Length of Stay (Days) and Avg Medications
by Primary Diagnosis Group



Executive Summary & Strategic Clinical Insights

Core Findings

High Inpatient Utilization (Primary Risk Driver)

- **Metric:** 3+ Prior Inpatient Visits -> **25.66% Readmission Rate**
- **Takeaway:** Frequent acute care utilization is a major predictor of 30-day return. Patients with 3+ prior visits readmit at more than double the 11.16% baseline.

High-Risk Post-Acute Discharge Pathways

- **Metric:** Transfers to Outpatient/Inpatient Facilities -> **36.69% Readmission Rate**
- **Takeaway:** Transitions of care to post-acute care and outpatient facilities represent severe vulnerability. Discharge coordination failure is a major contributor to early returns.

Complex Care & Extended Length of Stay

- **Metric:** Hospitalization >8 Days -> **13.37% Readmission Rate**
- **Takeaway:** Patient encounters requiring extended stays (>8 days) or high medication counts (>16 medications) show compounding complexity and higher relapse rates.

Medication Adjustment Vulnerability

- **Metric:** Regimen Changes -> **11.82% Readmission Rate**
- **Takeaway:** Patients undergoing active medication regimen adjustments during admission carry elevated risk, pointing to post-discharge prescription adherence gaps.

Operational Action Plan

Focus Area	Operational Initiative	Targeted Metric	Impact Goal
Complex Transitions	Mandatory 48-hour post-discharge telehealth check-in for SNF/Rehab transfers.	Discharge Disposition	Reduce post-acute return rate by 15%
High Utilizers	Enroll patients with ≥ 3 prior inpatient visits into dedicated nurse case management.	Prior Inpatient Visits	Lower 30-day return rate to $< 18\%$
Medication Reconciliation	Pharmacist-led discharge medication review for all patients with regimen changes.	Medication Change	Mitigate medication-related readmissions

Data Governance, Schema Architecture & Methodology

Data Provenance & Scope

- **Source Dataset**: De-identified Clinical Diabetes Inpatient Dataset (diabetic_data.csv).
- **Observation Period**: 10-year period (1999-2008) across 130 US hospitals.
- **Volume**: 101,766 total patient encounters.
- **Primary Outcome Variable**: 30-Day Readmission Flag (Readmitted Status = <30).

Data Model Architecture

- **Schema Pattern**: Star Schema centered on Fact_Encounters.
- **Relationships**: 1-to-many single-direction relationships (1 to *) linked via surrogate key integer columns:
    - Dim_AdmissionType[AdmissionTypeKey]
    - Dim_DischargeDisposition[DischargeDispositionKey]
    - Dim_AdmissionSource[AdmissionSourceKey]
- **DAX Layer**: Isolated _Measures table housing all explicit measures; all implicit visual summation disabled.

Data Cleaning & ETL Rules (Power Query UI)

- **Missing Value Handling**: Replaced missing/unknown string values (?, NULL) with standard categories ("Unknown", "Not Available").
- **Column Exclusions**: Removed high-null features with >40% missing values (weight, payer_code, medical_specialty).
- **Calculated Feature Bins**: Added custom conditional columns for:
    - Age Group ([0-10) through [90-100))
    - Length of Stay Band (1-3 Days, 4-7 Days, 8+ Days)
    - Prior Inpatient Visits Band (0 Visits, 1 Visits, 2 Visits, 3+ Visits)

Patient Privacy & Data Compliance

- **De-identification**: All Direct Identifiers (Patient Names, SSNs, exact admission dates) were not present in the source data.
- **Surrogate Identification**: Encounters and individual patients are referenced exclusively via de-identified IDs (Patient ID, Encounter ID).